ADS CMCE Practice Quiz 1

Total points 28/40

The respondent's email (**rushikesh.sambare.cmfeb26@gmail.com**) was recorded on submission of this form.

0 of 0 points

Name *

Rushikesh Sambare

12 Digit PRN *

260240320142

MCQ

28 of 40 points

✕ Which of the following operations is **not efficient in an ArrayList** compared to a LinkedList? *0/1

- ○ A) Random access by index
- ○ B) Iterating through elements
- ○ C) Inserting/removing in the middle
- ◉ D) Accessing the first element ✕

Correct answer

- ◉ C) Inserting/removing in the middle

✓ What is the output of the following code? * 1/1

```
int[] arr = {11, 22, 33, 44, 55};
int key = 100;
int index = -1;

for (int i = 0; i < arr.length; i++) {
   if (arr[i] == key) {
      index = i;
   }
}
System.out.println(index);
```

- ○ 100
- ○ 4
- ◉ -1 ✓
- ○ Key Not Found

✓ Which of the following is the **parent class of all exceptions** in Java? * 1/1

- ◉ A) Throwable ✓
- ○ B) Exception
- ○ C) RuntimeException
- ○ D) Error

✓ What is the **default initial capacity** of an ArrayList in Java? * 1/1

- ○ 0
- ○ 16
- ◉ 10 ✓
- ○ 5

✓ Binary Search can be categorized into which of the following? * 1/1

- ○ Greedy algorithm
- ○ Dynamic programming
- ○ Brute Force technique
- ◉ Divide and conquer ✓

```
class Node {
   int data;
   Node next;
   Node(int d) { data = d; }
}
public class Test {
   public static void main(String[] args) {
      Node head = new Node(10);
      head.next = new Node(20);
      Node newHead = new Node(5);
      newHead.next = head;
      head = newHead;
      System.out.println(head.data);
   }
}
```

*

1/1

- A) 10
- B) 5 ✓
- C) 20
- D) NullPointerException

Which of the following is **NOT true** about recursion in Java? *

1/1

- A) Every recursive function must have a base case.
- B) Recursion can lead to StackOverflowError if base case is missing.
- C) Recursion always executes faster than iteration. ✓
- D) Recursion can call itself with smaller sub-problems.

✓ The array is as follows: 1,2,3,6,8,10. Given that the number 17 is to be searched. At which call it tells that there is no such element ?(By using linear search(recursive) algorithm) *1/1

- ○ 5th Call
- ○ 17th Call
- ◉ 7th Call ✓
- ○ The function call itself infinite time

✗ Which of the following is **true**? * 0/1

- ○ OptionA) Linear search is faster for small datasets.
- ◉ B) Binary search is better for large, sorted datasets. ✗
- ○ C) Binary search requires random access.
- ○ D) All of the above. 1

Correct answer

- ◉ D) All of the above. 1

What is the output of following code * 1/1

```
class Fun1
{
public static void main(String abc[])
{
  int arr[] ={10,20,30,40,50};
  System.out.println(arr[5]);
}
}
```

- 50
- Compile time error
- ArrayIndexOutofBound Exception ✓
- Stack overflow

Which of the following is the correct way to declare a multidimensional array in Java? *1/1

- int[ ] arr;
- int arr[[ ]];
- int[ ][ ] arr; ✓
- int[[ ]] arr;

```
import java.util.*;
public class Main {
    public static void main(String[] args) {
        int[] arr = {5, 1, 3, 2, 4};
        Arrays.sort(arr);
        System.out.println(Arrays.binarySearch(arr, 3));
    }
}
```

*

0/1

- (●) 4 ✗
- ( ) 2
- ( ) 3
- ( ) Undefined

Correct answer

- (●) 2

Which of the following statements about singly linked list is **false**? *

0/1

- (●) A) Traversal is possible only in one direction. ✗
- ( ) B) Random access by index is O(1).
- ( ) C) Insertion at head is O(1).
- ( ) D) Deletion at a given key may require traversal.

Correct answer

- (●) B) Random access by index is O(1).

✓ What is the output of the following code snippet? * 1/1

```
public class Test {
   static int fun(int n) {
      if (n == 0) return 0;
      return n + fun(n - 1);
   }

   public static void main(String[] args) {
      System.out.println(fun(4));
   }
}
```

- ○ 4
- ○ 0
- ◉ 10 ✓
- ○ Comliation error

✓ Which of these data types is used by operating system to manage the Recursion in Java? *1/1

- ○ Array
- ○ LinkedList
- ◉ Stack ✓
- ○ Queue

```
import java.util.*;
public class Test {
    public static void main(String[] args) {
        int[] arr = {1, 2, 3, 4};
        int[] arr2 = arr;
        arr2[1] = 99;
        System.out.println(arr[1]);
    }
}
```

*

0/1

- 1
- 2 ✕
- 99
- Complietime Error

Correct answer

- 99

✓ What will this code print? * 1/1

```
int[] arr = {2, 4, 6, 8, 10};
int key = 8;
int index = -1;

for (int i = 0; i < arr.length; i++) {
    if (arr[i] == key) {
        index = i;
        break;
    }
}
System.out.println(index);
```

- ○ A) 2
- ◉ B) 3 ✓
- ○ C) 4
- ○ D) -1

✗ Which Java interface provides the root of the **collection hierarchy**? * 0/1

- ○ A) Iterable
- ◉ B) Collection ✗
- ○ C) List
- ○ D) Map

Correct answer

- ◉ A) Iterable

✓ The optimal data structure used to solve Tower of Hanoi is _____ * 1/1

- ○ Tree
- ○ Heap
- ○ Priority queue
- ◉ Stack ✓

✓ import java.util.*; * 1/1

```
import java.util.*;
public class Test {
   public static void main(String[] args) {
      ArrayList list = new ArrayList();
      list.add("Java");
      list.add(100);
      System.out.println(list.get(1));
   }
}
```

- ○ A) Java
- ◉ B) 100 ✓
- ○ C) Compilation error
- ○ D) ClassCastException

✓ Which collection guarantees that elements are **sorted in natural order**? * 1/1

- A) HashSet
- B) LinkedHashSet
- C) TreeSet ✓
- D) PriorityQueue

✓ Which of the following is **true about HashMap in Java**? * 1/1

- A) It allows one null key and multiple null values.
- B) It doesn't maintain any order of keys.
- C) Key lookups are average O(1).
- D) All of the above ✓

✓ What will the following code do? * 1/1

```
Node head = new Node(10);
 head.next = new Node(20);
head.next.next = new Node(30);
 head = head.next;
System.out.println(head.data);
```

- A) 10
- B) 20 ✓
- C) 30
- D) NullPointerException

```
import java.util.*;
public class Test {
    public static void main(String[] args) {
        HashMap<Integer, String> map = new HashMap<>();
        map.put(1, "A");
        map.put(2, "B");
        map.put(1, "C");
        System.out.println(map.get(1));
    }
}
```

*

0/1

- A
- C
- Error
- Null

Correct answer

- C

Which of the following statements about arrays in Java is **false**? *

1/1

- A) Arrays are objects.
- B) Array size can be changed after creation.
- C) An array can store primitive or objects.
- D) Arrays have a length property.

What is the **time complexity** of inserting a node at the **beginning** of a singly linked list? *1/1

- (●) A) O(1) ✓
- ( ) B) O(n)
- ( ) C) O(log n)
- ( ) D) O(n log n)

```
import java.util.*;
public class Test {
    public static void main(String[] args) {
        ArrayList<Integer> list = new ArrayList<>();
        list.add(1);
        list.add(2);
        list.add(3);
        list.remove(1);
        System.out.println(list);
    }
}
```

*

1/1

- ( ) A) [1, 2, 3]
- (●) B) [1, 3] ✓
- ( ) C) [2, 3]
- ( ) D) Compilation error

✓ Which keyword is used to **explicitly throw an exception**? * 1/1

- ○ A) try
- ○ B) catch
- ◉ C) throw ✓
- ○ D) throws

✗ Elements in an array are accessed _____ * 0/1

- ○ Randomly
- ○ Sequentially
- ○ exponentially
- ◉ logarithmically ✗

Correct answer

- ◉ Randomly

```
import java.util.*;
public class Test {
    public static void main(String[] args) {
        HashMap<Integer, String> map = new HashMap<>();
        map.put(null, "A");
        map.put(null, "B");
        System.out.println(map.size());
        System.out.println(map.get(null));
    }
}
```

*

0/1

- A) 2 and A
- B) 1 and B
- C) 2 and B
- D) Compilation error

Correct answer

- B) 1 and B

What is the **default initial capacity** of a HashMap? *

0/1

- A) 8
- B) 10
- C) 16
- D) 32

Correct answer

- C) 16

✓ Which of the following methods actually exists in ArrayList to check if an element is present? *1/1

- ○ A) has()
- ◉ B) contains() ✓
- ○ C) exists()
- ○ D) search()

✓ A linear collection of data elements where the linear node is given by means of pointer is called? *1/1

- ○ Array
- ○ Stack
- ○ Queue
- ◉ LinkedList ✓

What is the output of the following code? * 1/1

```
public class Test {
   public static void main(String[] args) {
      int[] arr = {10, 20, 30, 40};
      for (int i = 0; i < arr.length; i++) {
         arr[i] = arr[i] + i;
      }
      System.out.println(arr[2]);
   }
}
```

- 30
- 21
- 31
- 32 ✓

Which of the following conditions must be true for **Binary Search** to work correctly? *1/1

- A) Array must be sorted ✓
- B) Array must contain only integers
- C) Array size must be a power of 2
- D) Array must not contain duplicates

✕ What is output of the following code? * 0/1

```
public class Demo1 {
   static int sumDown(int n) {
      if (n == 0) return 0;
      return n + sumDown(n);
   }
   public static void main(String[] args) {
      System.out.println(sumDown(5));
   }
}
```

- (●) 25 ✕
- ( ) 20
- ( ) 5
- ( ) StackOverflowError

Correct answer

- (●) StackOverflowError

✓ What will be the output of the following code? * 1/1

```
public class Test {
   public static void main(String[] args) {
      int[] arr = new int[5];
      System.out.println(arr[2]);
   }
}
```

- (●) A) 0 ✓
- ( ) B) Garbage value
- ( ) C) Compilation error
- ( ) D) ArrayIndexOutOfBoundsException

✓ Which statement about ArrayList resizing is correct? * 1/1

- (●) A) It doubles its capacity every time it runs out of space. ✓
- ( ) B) It increases by 1 element each time.
- ( ) C) It triples its capacity.
- ( ) D) It stays fixed.

What is the **load factor** of a HashMap by default? * 1/1

- A) 0.5
- B) 0.65
- C) 0.75 ✓
- D) 1.0

```
public class Main{
    public static void main(String[] args) {
        int[][] arr = new int[2][];
        arr[0] = new int[]{1,2,3};
        arr[1] = new int[]{4,5};
        System.out.println(arr[1][2]);
    }
}
```

* 0/1

- A) 0
- B) 5 ✗
- C) ArrayIndexOutOfBoundsException
- D) Compilation error

Correct answer

- C) ArrayIndexOutOfBoundsException

This content is neither created nor endorsed by Google. - Contact form owner - Terms of Service - Privacy Policy

Does this form look suspicious? Report

Google Forms